SETLIST

START HERE | NOW | HUM QUALITY | AGENT AND RELIABILITY | VALIDATION CORRECTNESS | RETRIEVAL AND DATA | TESTS AND HOUSEKEEPING | LATER | SHIPPED | IDEAS

NEXT STEPS

StrudelAssistant Setlist

What comes next for the web app, now that the iOS shell is gone. The branch is merged, credits are in, and the app is on Luna with a model picker. The step that matters most now is recording takes, and the rest can be handed off in almost any order.

merged to **main** at **e084dfa** **27** steps · **6** ideas · **9** done · **3** need you 15 October 2026

START HERE

Record a take and look at what it heard

Every accuracy number so far comes from synthesized audio. A recorded take is the one test that can overturn the tuning, and it takes fifteen minutes. Playing a pattern on strudel.cc into the Hum button works well because you know the source; a human voice is a separate test.

1. Open the page in Safari or Chrome. Embedded previews block the microphone.
2. Press the red dot, play or hum, press it again.
3. Judge the panel labelled *As hummed* first. It involves no model. Use Download recording to keep the take.

```
# in the StrudelAssistant conda env
python app/server.py
# then open http://localhost:8000
```

For each take, note three things (and which source pattern played, if any): did held notes shatter into repeats, did repeated notes merge into one, and was the tempo right, half or double.

NOW	HUM QUALITY	AGENT AND RELIABILITY	VALIDATION CORRECTNESS
3 steps	4 steps	5 steps	3 steps
RETRIEVAL AND DATA	**TESTS AND HOUSEKEEPING**	**LATER**	**IDEAS**
4 steps	5 steps	3 steps	6, not commitments

NOW

Blocked on you

Credits and the Luna switch are done. What is left here is recording takes, because every accuracy number so far comes from synthesized audio.

01 ○ DONE done 15 Oct

Add OpenAI credits

The account returned `insufficient_quota` on 14 October, so nothing that calls the API could be tested. Credits were added on 15 October.

RESULT

Tool-calling requests succeed, and live `/api/chat` and `/api/hum` requests returned valid answers on the new model.

02 ○ DONE done 15 Oct

Switch to `gpt-5.6-luna`

Luna scored 5/5 on the melody benchmark, the same as `gpt-5.4-mini`, at $0.20 / $1.20 per million tokens against $0.75 / $4.50. It needs `reasoning_effort="none"` to use function tools on Chat Completions and still returns string content.

RESULT

`core.py` now uses it and `docs/decisions.md` records the benchmark. The benchmark covers one melody task, so plain Strudel questions and hard multi-step fixes are unmeasured. Raise `reasoning_effort` if answers get worse.

03 ● YOU 15 min

Record real takes: a pattern on strudel.cc, and a human voice

The transcription was tuned on synthesized hums. Your usual test, playing a pattern on strudel.cc into the Hum button, gives ground truth because the source pattern is known, but it is synth audio through speakers and a room, not a voice. A human voice is still its own test.

HOW	Record a few of each: a slow pattern with repeated notes, a fast one, and a tune you know well. Keep low notes above C2, because the transcriber's floor is C2 and laptop speakers often do not reproduce lower ones. Reverb (`.room`) smears note boundaries, so expect it to be harder. Use Download recording to keep each take.
DONE WHEN	For each take you can say whether held notes shattered, whether repeated notes merged, and whether the tempo was right, half or double.
UNBLOCKS	04 and 05

HUM QUALITY

Making the transcription trustworthy

Everything here is measured against real takes, so it waits on step 03.

04 ○ HAND OFF small

Turn your takes into regression tests

Synthesized audio proves the pipeline works. Only labelled real recordings prove it keeps working on them.

HOW	Save five to ten takes from Download recording, each with the notes you meant (or the strudel.cc pattern that played) written beside it, as fixtures next to the realistic suite. The expected notes must come from you, not from the transcription, or the test only checks the code against itself. Keep recordings of your voice in an ignored folder if the repository will be public. Allowing one wrong note per take is likely more realistic than demanding an exact match.
DONE WHEN	`pytest` runs them alongside the existing 64 tests.
DEPENDS ON	03

05 ○ HAND OFF small to medium

Tune transcription on your takes

The loudness-dip prominence, the 80 ms minimum note and the 60–180 bpm tempo window were all set on synthesized audio.

DONE WHEN	Your recorded fixtures pass, and so do the synthetic tests.
DEPENDS ON	04, so every change is measured rather than judged by ear

06 ○ HAND OFF small

Fix tempo in one click

Half- and double-time readings are inherent to hearing a pulse. Correcting one today means editing the `setcpm` line by hand.

HOW	Add ×2 and ÷2 controls to the As hummed panel that rewrite `setcpm` and replay.
DONE WHEN	A take read at half tempo can be corrected without typing.

07 ○ DONE done 15 Oct

Settle the tempo numbers and count the fallback

`_estimate_tempo`'s docstring says 19 of 21 realistic takes land within 10%, while the commit and `docs/humming.md` say 21 of 21. The `melody.py` docstring and the humming diagram still say tempo comes from `beat_track`. Nothing counts how often the `beat_track` fallback fires.

RESULT	Re-measured on all 21 realistic takes: the median-gap method is 21 of 21 within 10% (also 21 of 21 within 5%) and `beat_track` alone is 8 of 21. The docstring, diagram and docs are corrected, and the fallback now prints a warning each time it fires, with two new tests.

AGENT AND RELIABILITY

Fewer failures, lower latency

Independent of each other. Ordered by how soon they pay off.

08 ○ DONE done 15 Oct

Retry model calls with backoff

A rate limit used to fail the whole request after two quick attempts. The hum endpoint still returned its transcription, but the chat endpoint returned only an error.

RESULT — The chat model now allows 5 retries with exponential backoff, about 15 seconds of waiting. An out-of-credits error now says so instead of calling it a rate limit. The retry path itself is untested, because it needs a real 429.

09 ○ CLOSED closed 15 Oct

Build the agent once

`run_llm` builds a new agent on every request. Measured: that takes about 1.2 ms (max 2.7 ms over 20 builds), against roughly 10 s for the model call.

RESULT — Not worth changing. The comments and docs that called it wasteful are corrected. Revisit only if the ratio changes.

10 ○ HAND OFF medium

Keep a warm sandbox

Every validation starts Node and imports Strudel from scratch, and a correction loop pays that two or three times per answer.

TRADE-OFF — A long-lived worker gives up per-request isolation. Restart it every few dozen patterns and after any timeout, and keep the same permission flags.

DONE WHEN — Validation latency is measured before and after, and the file-write, network and subprocess escapes still fail inside the worker.

11 ○ DONE done 15 Oct

Say so when the sandbox is missing

If `node` is not on the path or `sandbox/node_modules` is missing, the tool tells the agent to answer unverified and the server used to log nothing.

RESULT — `core.py` prints a "Pattern validator disabled" warning once at startup, and the validator and the warning share one check. Verified by temporarily moving `node_modules` aside.

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12 ○ HAND OFF small

Support the Responses API

`gpt-6-astra` and `gpt-5.6-sol` reject function tools on Chat Completions, and their replies arrive as content blocks that `run_llm` and the server treat as strings. Sol turned out not to need it: with `reasoning_effort="none"` it works on Chat Completions and is in the model picker. Only `gpt-6-astra` still needs this.

DONE WHEN — `gpt-6-astra` runs through the full pipeline. Re-benchmark before offering it, since it did not beat the cheaper models on this task.

VALIDATION CORRECTNESS

Making VALID mean it plays

Three gaps between what the sandbox accepts and what the browser plays.

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13 ○ HAND OFF small

Check sound names against the real sample list

The validator flags `gm_` soundfonts by prefix only. An invented drum name passes validation and then plays silence.

HOW — Fetch the dirt-samples sample map once, at install time, and ship it with the sandbox. The sandbox has no network by design, so it cannot fetch the list when it runs.

DONE WHEN — A pattern using `sound("bdd")` comes back with a warning naming the unknown sound.

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14 ○ HAND OFF small to medium

Align the two Strudel versions

The page plays with `@strudel/web` 1.3.0, and the sandbox validates with `@strudel/core` 1.2.6. A function that exists in only one of them is judged by the other.

HOW	Find which core release the page's bundle contains. Then pin the page to a release that matches the sandbox, or run the sandbox on the page's own bundle if it will load in Node.
DONE WHEN	Both sides report the same version, and a test fails if they ever diverge.

15 ○ HAND OFF tiny

Revisit the `@kabelsalat/web` stub

The stub exists because every `@kabelsalat/web` release from 0.3.0 to 0.4.1 lacks `SalatRepl` and `@strudel/core` has nothing newer than 1.2.6.

DONE WHEN	Either a newer core or a fixed `@kabelsalat/web` is out and the stub is removed, or the check is repeated and the stub stays.

RETRIEVAL AND DATA

Keeping the corpus honest

BM25 and Pinecone are two views of one corpus, and they drift.

16 ○ HAND OFF small

Stop Pinecone drifting from the chunk dump

Ingestion overwrites `chunks.json` but upserts vectors, so a changed chunk leaves its old vector behind to keep surfacing in results.

DONE WHEN	Ingestion deletes vector ids that no longer appear in `chunks.json`.

17 ○ HAND OFF tiny

Decide whether `data/chunks.json` is committed

It is gitignored, but the README, `docs/extending.md` and the architecture traps say to commit it. A fresh clone has no dump, so retrieval quietly falls back to vector-only.

DONE WHEN Either the file is committed and the ignore rule removed, or every doc says it must be generated first.

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18 ○ **DONE** done 15 Oct

Make a failed ingestion batch fail loudly

`add_batch` logged each batch's exception itself, so the closing success line printed and the process exited 0 even when batches failed. `chunks.json` is written first, so BM25 ends up ahead of Pinecone.

RESULT A failed batch now ends the run with the failed batch numbers and exit code 1. Checked against a stubbed Pinecone with no failures, two failures and all failing, not a real failed upload, which would be slow and paid.

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19 ○ **HAND OFF** medium

Tune the hybrid weights on real queries

The 50/50 split and `k=10` per retriever are a reasonable default, not a measured one.

DONE WHEN There is a small set of real questions and the weights are chosen against it. Needs usage data, so it waits on 33.

TESTS AND HOUSEKEEPING

Smaller things that add up

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20 ○ **PARTLY DONE** partly done 15 Oct

Add tests beyond `melody.py`

The server's HTTP layer is now covered by 26 tests in `tests/test_server.py`, with `run_llm` stubbed so nothing touches the network or `backend.core`: model validation, history filtering, HTML escaping, source de-duplication, error messages, and the hum

endpoint's 400, 413, 422 and degrade-to-200 paths. Four deliberate bugs in `server.py` each failed exactly one test.

STILL UNTESTED	The agent loop, retrieval, the sandbox report parser in `validate_pattern`, and the frontend (including the model picker's JavaScript).
DONE WHEN	The sandbox parser and the picker are covered, and the agent and retrieval each have at least a smoke test.

21 ○ HAND OFF tiny

Fix `.env.example`

It describes an `OPENAI_CHAT_MODEL` variable and a `search_web` tool that do not exist, and marks `PINECONE_API_KEY` optional when the app needs it.

DONE WHEN	It matches what the code reads.

22 ○ DONE done 15 Oct

Fix the ingestion docstring

The docs said `ingestion.py`'s module docstring claims it crawls `python.langchain.com`.

RESULT	The docstring already described `strudel.cc` correctly. Only the stale "known rough edges" bullet in `docs/architecture.md` was wrong, and it is removed.

23 ● YOU 15 min

Confirm or delete the inferred decisions

Several entries in `docs/decisions.md` are marked "inferred — confirm": hybrid retrieval weights, FastAPI over Flask, Pinecone over Chroma, and the vanilla frontend over Gradio.

DONE WHEN	Each is confirmed with its real reason, or deleted.

24 ○ DONE done 15 Oct

Review and merge the branch

The hum, agent and model-picker work sat on `feat/hum-to-pattern`, 24 commits ahead of `main`.

RESULT — Merged as pull request #1, followed by #2 for the copy button. `origin` now points at `zulle09/Strudel-AI-Assistant`.

LATER

Before anyone else uses it

None of these matter while the app runs on your own machine. All of them matter the day it does not.

25 ○ HAND OFF medium

Pick a host and serve over HTTPS

`python app/server.py` binds localhost only and nothing in the repo says where it would run. `docs/runbook.md` has a TODO where the deploy steps should be.

DONE WHEN — The backend has a real hostname and certificate, and the runbook has real deploy, rollback and what-to-check-when-it's-down steps.

26 ● YOU a conversation with a lawyer

Get the AGPL question answered

Strudel is AGPL-3.0. The sandbox runs it inside a networked service, which is the case the licence's network clause addresses.

DONE WHEN — You know what source you must publish.

27 ○ HAND OFF small

Add rate limits and a spend cap

Both endpoints call paid APIs and neither is limited, so an open instance is an open tap on the OpenAI bill. The model picker makes it worse: any visitor can choose Sol, the priciest model.

DONE WHEN

Requests are limited per client and a daily spend ceiling stops the model calls. Consider restricting the picker or defaulting visitors to Luna.

SHIPPED

Done, but never a numbered step

Model picker

A dropdown above the composer chooses between GPT-5.6 Luna (default), Terra and Sol. The server only accepts ids in `app/backend/models.py`. Terra and Sol have had one live answer each, not the melody benchmark.

Conda instead of uv

The project runs from the `StrudelAssistant` conda env, described by `environment.yml`. There is no lockfile, so installs are not reproducible.

Copy message

A button on each answer copies the reply as markdown. The patterns inside keep their own Copy button.

Download recording

A hum result can save the 16 kHz WAV the server analysed, so real takes can become test fixtures (step 04).

Server tests

26 tests cover the HTTP layer with a stubbed model. Part of step 20.

Web only

The iOS shell and CORS were removed on 14 October. The shell is recoverable from commit `2d88161`.

IDEAS

Worth exploring, not yet worth planning

28 · Lyrics from audio

29 · Spoken instructions

Record or attach audio and ask about the lyrics. Needs a speech-to-text step, because today the agent never hears the audio and `/api/chat` takes text only. The text would feed the chat flow beside the melody summary.

Audio models failed at pitch, but transcribing words is what they are good at. Saying "make it darker" is a natural gesture. Shares the speech-to-text step with 28.

30 · Chords and harmony

The pipeline is monophonic. A polyphonic transcriber such as Spotify's basic-pitch could add harmony, if its dependencies install on Python 3.13.

31 · Swing and triplets

The grid is straight sixteenth notes, so a shuffle is forced onto the nearest step.

32 · A native app again

The Capacitor shell was removed on 14 October to focus on the web app. It is recoverable from git history (`2d88161`) and only makes sense once the backend has a deployment target.

33 · Measure it in use

LangSmith tracing is already wired through environment variables, but nothing records latency, tool calls per answer, or how often validation changes the model's pattern.

Grounded in the code at commit `e084dfa`. For how the pieces fit together, see docs/architecture.md.